

1. There are 3 variables. so we get

$$0 = 000$$

$$1 = 001$$

$$2 = 010 \text{ — connected to OR gate}$$

$$3 = 011$$

$$4 = 100 \text{ — connected to OR gate}$$

$$5 = 101 \text{ — connected to OR gate}$$

$$6 = 110$$

$$7 = 111$$

Canonical Sum-of-products

$$F(x, y, z) = 010 + 100 + 101$$

$$F(x, y, z) = x'yz' + xy'z' + xy'z$$

$$1 = a$$

$$0 = a'$$

Canonical Product-of-Sums

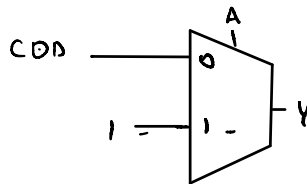
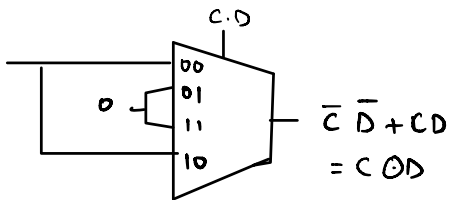
$$F(x, y, z) = (0+0+0) * (0+0+1) * (0+1+1) * (1+1+0) * (1+1+1)$$

$$1 = a'$$

$$0 = a$$

$$F(x, y, z) = (x+y+z) * (x+y+z') * (x+y'+z') * (x'+y'+z) * (x'+y'+z')$$

2.



$$Y = \bar{A}(CD) + A \cdot 1$$

$$= \bar{A}(\bar{C}\bar{D} + CD) + A$$

$$Y = \bar{A}\bar{C}\bar{D} + \bar{A}CD + A$$

3. I have made the truth in word and then added it here.

D3	D2	D1	D0	Sa	Sb	Sc	Sd	Se	Sf	Sg
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	1	0	1
0	0	1	1	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	1	0	1	1	0	1	1
0	1	1	0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	0	0	1	1
1	0	1	0	0	0	0	0	0	0	0
1	0	1	1	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0	0	0	0
1	1	0	1	0	0	0	0	0	0	0
1	1	1	0	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0

b. Simplified Boolean functions for output

For $S_a = \bar{D}_2 \bar{D}_1 \bar{D}_0 + D_3 \bar{D}_2 \bar{D}_1 + \bar{D}_3 D_2 D_0 + \bar{D}_3 D_1$

For $S_b = \bar{D}_3 \bar{D}_1 \bar{D}_0 + \bar{D}_3 D_1 D_0 + \bar{D}_3 \bar{D}_2 + \bar{D}_2 \bar{D}_1$

For $S_c = \bar{D}_2 \bar{D}_1 + \bar{D}_3 D_0 + \bar{D}_3 D_2$

For $S_d = \bar{D}_3 D_2 D_1 D_0 + \bar{D}_2 \bar{D}_1 \bar{D}_0 + \bar{D}_2 \bar{D}_2 D_1 + \bar{D}_3 D_2 \bar{D}_0$

For $S_e = \bar{D}_3 D_1 \bar{D}_0 + \bar{D}_2 \bar{D}_1 \bar{D}_0$

For $S_f = \bar{D}_2 \bar{D}_1 \bar{D}_0 + \bar{D}_3 D_2 \bar{D}_1 + \bar{D}_3 D_2 \bar{D}_0 + D_3 \bar{D}_2 \bar{D}_1$

For $S_g = \bar{D}_3 \bar{D}_2 D_0 + \bar{D}_3 D_2 \bar{D}_1 + \bar{D}_3 D_2 \bar{D}_0 + D_3 \bar{D}_2 \bar{D}_1$

4. Using the given circuit

$$Y = \bar{A}D + A\bar{C}D + A\bar{B}C + ABCD$$

$$Z = BD + A\bar{C}D$$

Canonical Sum-of-Products for Y

$$= \bar{A}BD + \bar{A}\bar{B}D + AB\bar{C}D + A\bar{B}\bar{C}D + A\bar{B}CD + A\bar{B}\bar{C}\bar{D} + ABCD$$

$$= \bar{A}B\bar{C}D + \bar{A}BCD + \bar{A}\bar{B}CD + \bar{A}\bar{B}\bar{C}D + AB\bar{C}D + A\bar{B}\bar{C}D + A\bar{B}CD + A\bar{B}\bar{C}\bar{D} + ABCD$$

$$= \sum (m_5, m_7, m_3, m_1, m_{13}, m_9, m_{11}, m_{10}, m_{15})$$

Canonical Sum-of-Products for Z

$$= ABD + \bar{A}BD + AB\bar{C}D + A\bar{B}\bar{C}D$$

$$= ABCD + AB\bar{C}D + \bar{A}BCD + \bar{A}\bar{B}\bar{C}D + AB\bar{C}D + A\bar{B}\bar{C}\bar{D}$$

$$= ABCD + AB\bar{C}D + \bar{A}BCD + \bar{A}\bar{B}\bar{C}D + A\bar{B}\bar{C}\bar{D}$$

$$= \sum (m_{15}, m_{13}, m_7, m_5, m_9)$$

5. From Question 4.

K-map for Y = $\sum (5, 7, 3, 1, 13, 9, 11, 10, 15)$

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$	0	1	3	2
$\bar{A}B$	4	5	7	6
AB	12	13	15	14
$A\bar{B}$	8	9	11	10

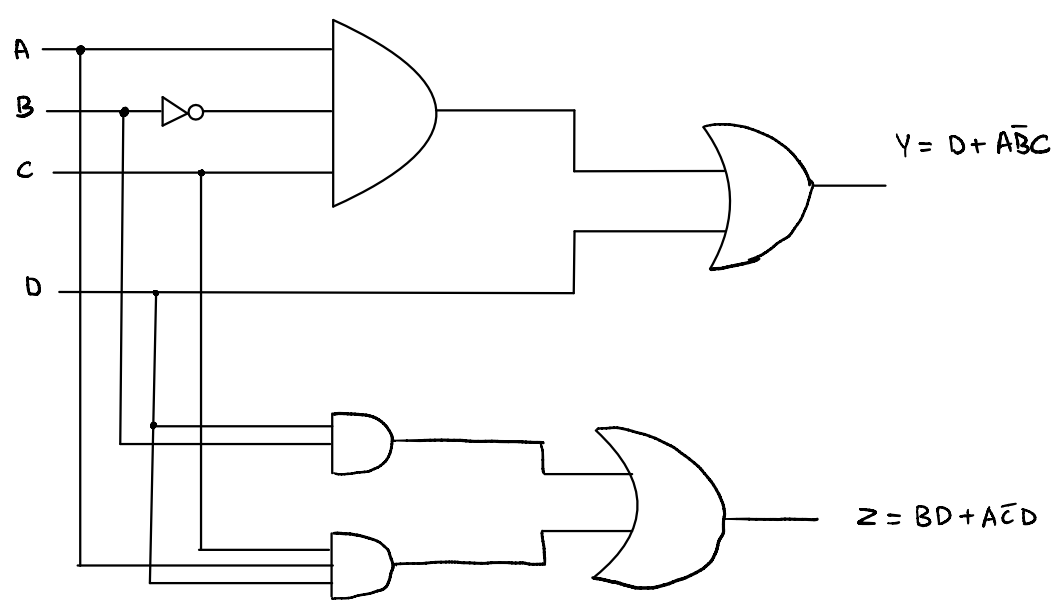
K-map for Z = $\sum (15, 13, 7, 5, 9)$

AB \ CD	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$
$\bar{A}\bar{B}$		1	1	
$\bar{A}B$		1	1	
AB		1	1	
$A\bar{B}$		1	1	1

$$Y = D + A\bar{B}C$$

$$Z = BD + A\bar{C}D$$

Improved circuit is as follows.



6. Took a screenshot from the pdf.

X	$Q_A(t)$	$Q_B(t)$	Next State	
			$Q_A(t+1)$	$Q_B(t+1)$
0	0	0	1	0
1	0	0	0	0
0	0	1	0	1
1	0	1	1	1
0	1	0	1	0
1	1	0	1	0
0	1	1	1	1
1	1	1	0	1